

Narrative Crystallization: A Deterministic Framework for Measuring Global Narrative States

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Abstract

Human societies interpret events primarily through narrative structures rather than isolated data points. This paper proposes **Narrative Quantification**, a framework for transforming narrative information into structured cognitive data. By decomposing narratives into actors, events, conflicts, resolutions, emotional context, and causal relationships, narratives can be stored as structured datasets and analyzed computationally.

To operationalize this transformation, the paper introduces the concept of a **Narrative Compiler**, a system that converts narrative text into structured cognitive representations such as the **24TAG structure**. These representations enable the creation of narrative databases that may serve as the foundation for **Large Cognitive Models (LCM)**—systems designed to reason over structured representations of human narratives rather than purely probabilistic language tokens.

This work suggests that narrative quantification may function as a new layer of cognitive infrastructure for future AI systems.

This framework introduces a measurement layer for global narratives, enabling the construction of a time-evolving narrative state space.

1 Introduction

Human cognition is strongly structured by narrative. Cognitive research suggests that humans organize experiences not merely as isolated facts but as meaningful sequences of events forming narrative structures. According to Jerome Bruner [Bruner, 1990], narrative provides one of the fundamental modes through which humans interpret reality. Similarly, Roger Schank and Robert Abelson [Schank and Abelson, 1977] proposed that human knowledge is stored in structured scripts describing typical sequences of events. Communication theory further characterizes humans as *homo narrans*, a species that understands and communicates meaning through stories [Fisher, 1984]. These perspectives suggest that narrative structures are deeply embedded in human cognition and memory.

The idea of augmenting human knowledge through structured information systems has a long intellectual history. Vannevar Bush [Bush, 1945] envisioned the Memex as a system that would allow humans to navigate associative trails of knowledge. Later, Douglas Engelbart [Engelbart, 1962] proposed computer systems designed to augment human intellect by organizing and manipulating complex information structures. These early visions influenced the development of the modern World Wide Web and subsequent attempts to structure knowledge for machine interpretation. The Semantic Web proposed by Tim Berners-Lee and colleagues [Berners-Lee et al., 2001] represents a major effort to represent knowledge in machine-readable forms using structured data and linked resources.

Recent advances in artificial intelligence have further expanded the ability of machines to process large-scale textual information. Neural language models represent semantic relationships in high-dimensional vector spaces, enabling systems to capture statistical patterns of language [Mikolov et al., 2013]. At the same time, hybrid architectures combining neural models with external knowledge retrieval have become increasingly important. Retrieval-Augmented Generation (RAG) integrates language models with database search mechanisms, allowing AI systems to retrieve relevant knowledge during inference [Lewis et al., 2020]. From the perspective of information theory, information processing can be interpreted as the reduction of uncertainty, a concept originally formalized by Claude Shannon [Shannon, 1948]. In this context, knowledge retrieval can be viewed as a process that reduces informational entropy by selecting relevant structures from a larger information space.

Research in artificial intelligence increasingly explores cognitive inspirations for machine intelligence. Work by Yoshua Bengio [Bengio, 2017] investigates mechanisms that might enable machine learning systems to form structured representations resembling aspects of conscious reasoning. Similarly, research discussed by Demis Hassabis and colleagues [Hassabis et al., 2017] emphasizes the importance of integrating insights from neuroscience into artificial intelligence architectures.

Despite these advances, most current systems still treat information primarily as static documents rather than structured event-based narratives. This paper proposes the **Narrative Quantification** framework, in which textual information is decomposed into narrative event structures that can be represented, stored, and retrieved as part of a computable narrative knowledge space. In this framework, information retrieval can be interpreted as an entropy-reduction process over a structured narrative space, and the resulting crystallized representations serve as the foundation for **Large Cognitive Models (LCM)**—systems capable of reasoning over structured narrative representations rather than purely probabilistic language tokens.

2 Narrative Cognition

Human cognition frequently follows a narrative processing loop:

$$\text{Event} \rightarrow \text{Meaning} \rightarrow \text{Narrative} \rightarrow \text{Memory}$$

Experiences are rarely stored as raw data. Instead, individuals interpret events through causal explanations and emotional context, forming narratives that are later stored in memory. This mechanism allows humans to compress complex experiences into structured stories that predict future outcomes and guide subsequent decisions. Narrative cognition therefore represents a fundamental cognitive process rather than a mere cultural artifact.

3 Narrative Quantification

Narrative Quantification is the process of converting narrative structures into analyzable data. Narratives can be decomposed into structural components: Actors, Events, Conflict, Resolution, Emotional context, and Causal relationships. Once extracted, these components can be represented through structured tags and stored within databases.

3.1 Decomposition

The first stage of quantification is **Decomposition**, where raw narrative text is segmented into its primary semantic units—**Actors**, **Events**, and **Causal Relationships**—transforming linguistic ambiguity into a set of distinct variables.

3.2 Event Formation

The second stage is **Event Formation**, where continuous occurrences are discretized into logical structures. An *event* is defined as the minimum causal interaction unit between actors unfolding over time. A typical event unit is structured as:

- **Actors:** The agents initiating or receiving the interaction.
- **Action–Reaction Pair:** The core causal loop (e.g., Actor A exerts influence, Actor B retaliates).
- **Temporal Boundary:** The specific window from the trigger to the resulting outcome.

3.3 Narrative Compression

The final stage is **Narrative Compression**, an active entropy reduction within the information space. By applying decomposition and event formation, irrelevant details are filtered out and only the causal, emotional, and structural core is retained:

- **Computational Efficiency:** Reasoning over discrete, compressed event units is far more efficient than processing raw linguistic tokens.
- **Long-Term Coherence:** The narrative arc becomes visible, enabling tracking of long-term dynamics.
- **Storage and Retrieval:** Compression enables the construction of vast, searchable Narrative Databases.

3.4 Observational Projection: Transforming 5W1H into Event Coordinates

The Narrative Quantification framework introduces an **Observational Projection**, converting descriptive observations into structured **Event Coordinates**.

3.4.1 Observation Representation

Let an observation extracted from text be defined as:

$$O = (who, what, when, where, why, how)$$

We define a transformation $\Phi : O \rightarrow E$, where E represents the Event Coordinate.

3.4.2 Event Coordinate Structure

An event is decomposed into multiple structured components:

$$E = (A, R, T, S, C, M)$$

where A is the Actor set, R the Actor relationships, T the Temporal vector, S the Spatial vector, C the Causal classification, and M the Mechanism category.

3.4.3 Temporal Expansion

Natural language typically expresses time as a single timestamp. The system expands temporal information into a **temporal vector**:

$$T = (t_{\text{cause}}, t_{\text{event}}, t_{\text{impact}})$$

corresponding to the `causality_direction` dimension (TAG08) in the 24TAG structure.

3.4.4 Spatial Expansion

The spatial coordinate is represented as:

$$S = (s_{\text{origin}}, s_{\text{event}}, s_{\text{impact}})$$

allowing a single event to propagate across spatial domains while remaining within the same causal unit.

3.4.5 Actor and Relationship Structure

Actors are represented as a set $A = \{a_1, a_2, \dots, a_n\}$. Interactions are represented as typed directed relations:

$$a_i \xrightarrow{[\tau]} a_j, \quad \tau \in \{\text{cooperate}, \text{oppose}, \text{sanction}, \text{respond}, \dots\}$$

per the predicate classification defined in TAG03.

3.4.6 Implementation of the Projection Function

In practice, Φ is not a deterministic function but a **probabilistic estimator**:

$$\hat{E} = \Phi(O; \theta)$$

where θ represents the parameters of the extraction model. The transformation proceeds in three stages:

(1) **Entity Extraction**, (2) **Semantic Classification**, and (3) **Constraint Filtering**.

3.4.7 Causal Classification

The component C represents a causal classification *inferred from temporal precedence and interaction structure*. The system derives causal direction from the temporal ordering of $T = (t_{\text{cause}}, t_{\text{event}}, t_{\text{impact}})$ and the interaction pattern encoded in R .

3.4.8 Integration with Narrative Graph Construction

The resulting event coordinate $\hat{E}$ forms the atomic unit used in the Narrative Graph construction described in Section 4. Each event coordinate becomes a node, while relations encoded in R , T , and C define the edges.

4 The Narrative Compiler: Search and Optimization

The Narrative Compiler explores a vast space of potential interpretations to find the most probable narrative structures, generating the final probability distribution $P(N \mid \text{text})$. Figure 1 raw news §0–§3 “Pydantic RAG 0%

4.1 Generating Possible Narrative Graphs

The Compiler treats narrative construction as a graph-search problem, building a **Narrative Graph** G where nodes represent events and edges represent causal or temporal transitions.

4.2 Beam Search for Hypothesis Management

To maintain a manageable set of narrative states, the Compiler utilizes **Beam Search**, keeping only the top- k most plausible narrative hypotheses at each step.

4.3 Graph Pruning and Information Entropy

Graph Pruning is the active removal of low-probability or redundant branches:

- **Redundancy Filtering:** Multiple paths leading to the same structural state are pruned.
- **Entropy Thresholding:** Branches increasing conditional entropy without semantic value are eliminated.

4.4 Causal Constraints: The Logical Filter

The Narrative Compiler applies logical rules to validate every proposed edge:

- **Temporal Precedence:** A cause must strictly precede its effect.
- **Actor Consistency:** Actions must align with actor capabilities and motivations.
- **Interaction Logic:** The Action–Reaction loop must maintain coherent flow.

4.4.1 Narrative Graph Scoring Function

The constraints above are integrated into a unified **scoring function**:

$$S(G) = \alpha \cdot \Gamma_{\text{causal}} + \beta \cdot \Lambda_{\text{consistency}} + \gamma \cdot \Omega_{\text{compression}}$$

where

$$\Gamma_{\text{causal}} = \sum_{(E_i \rightarrow E_j) \in G} C_{ij}, \quad \Lambda_{\text{consistency}} = \sum_k L_k(G), \quad \Omega_{\text{compression}} = -(|V| + |E|)$$

and α, β, γ are hyperparameters. Beam Search retains the top- k graphs ranked by $S(G)$ at each step.

4.5 Final State Synthesis

After iterative graph expansion, the Compiler selects the highest-scoring graph:

$$G^* = \arg \max_{G \in \mathcal{B}} S(G)$$

The selected graph is transformed into a **Narrative State Vector**:

$$\mathbf{s} = (T_1, T_2, \dots, T_{24})$$

and converted into a cryptographic fingerprint:

$$h = H(\mathbf{s})$$

where H is a cryptographic hash function.

4.6 Narrative State Representation

Once event coordinates are connected into a Narrative Graph $G = (V, E)$, the system derives:

$$N = \Psi(G) = \Psi(V, E, \tau)$$

where τ represents the relation types defined in TAG03. This formulation establishes the narrative state as the direct mathematical antecedent of the `state_hash`.

5 The 24TAG Taxonomy

To ensure universal compatibility and deterministic hashing, the Narrative Quantification protocol defines a fixed schema of 24 structured tags categorized into four primary layers. Figure 2 24TAG JSON response_schema LLM T06T02

Rationale for the 24-Tag Schema

The schema is motivated by two considerations. First, **cognitive inspiration**: the tag set is designed with reference to the approximate complexity threshold that human cognition can integrate when interpreting causal narratives. Second, **empirical optimization**: the number and structure of tags were informed by narrative decomposition experiments aiming to maximize agreement in event interpretation. The 24TAG schema should not be interpreted as a fixed universal ontology but as an initial structural basis, extensible depending on domain, language, and cultural context.

Table 1: The 24TAG Framework Definitions

Layer	Tag ID	Tag Name	Description
Actor & Agency	T01	Primary_Actor	Main entity initiating the interaction
	T02	Secondary_Actor	Entity receiving or responding to the action
	T03	Actor_Role_A	Functional role of the primary actor
	T04	Actor_Role_B	Functional role of the secondary actor
	T05	Actor_Motivation_A	Underlying goal for Actor A
	T06	Actor_Motivation_B	Underlying goal for Actor B
Event & Action	T07	Action_Type	Category of the event
	T08	Action_Intensity	Magnitude of the action (0.0–1.0)
	T09	Target_Resource	Object or value being contested
	T10	Event_Modality	Factual, hypothetical, or desired
	T11	Temporal_Sequence	Relative order within the narrative arc
	T12	Spatial_Context	Domain or environment of the event
Causal & Logic	T13	Causal_Link_Type	Nature of the causal connection
	T14	Conflict_Nature	Type of opposition
	T15	Conflict_Intensity	Severity of friction between actors
	T16	Resolution_Status	Degree to which conflict has been settled
	T17	Outcome_Valence	Positive or negative impact on narrative state
	T18	Logic_Consistency	Alignment with prior narrative states
Context & Tone	T19	Emotional_Tone_A	Emotional state of Actor A
	T20	Emotional_Tone_B	Emotional state of Actor B
	T21	Perspective_Bias	Point of view from which narrative is told
	T22	Info_Asymmetry	Knowledge gap between actors
	T23	Significance_Score	Relative importance of event to overall arc
	T24	Narrative_Closure	Extent to which event concludes a thread

5.1 Data Structure and Compilation

Each tag ($T_{01...24}$) is populated by the Narrative Compiler. The `state_hash` is calculated by applying SHA-256 to the JCS-normalized string of these 24 tags:

$$\text{state_hash} = \text{SHA256}(\text{JCS}(\{TAG_{01...24}\} \setminus \{\text{state_hash}\}))$$

5.2 Integrity and Verification

By excluding the `state_hash` from its own input, we create a **One-Way Cognitive Seal**. A receiver can verify integrity by: (1) removing the `state_hash` value, (2) re-running JCS and SHA-256 on the remaining 24 tags, and (3) comparing the result with the original field. Any divergence indicates the narrative structure has been modified post-compilation.

6 Geometric Definition of Narrative Space

6.1 Narrative Space (Manifold)

We define an embedding function $e : G \rightarrow \mathbb{R}^d$ mapping each narrative graph into a d -dimensional vector space. Each narrative state corresponds to a point on the **narrative manifold**:

$$N = e(G) \in \mathbb{R}^d$$

6.2 Narrative Trajectories

As new events occur and interpretations evolve, narrative graphs generate **narrative trajectories**:

$$\mathcal{T} = \{e(G_t) \mid t = 1, 2, \dots, T\}$$

6.3 Quantifying Narrative Divergence

Differences between interpretations can be quantified as:

$$d(G_1, G_2) = \|e(G_1) - e(G_2)\|_2$$

Alternative metrics (cosine similarity, geodesic distances) may also be employed.

6.4 Interpretation Geometry

This geometric formulation enables: measuring divergence between interpretations, identifying clusters of similar narratives, detecting anomalous narrative trajectories, and analyzing convergence or polarization across populations.

7 Cognitive Infrastructure

7.1 From Knowledge Bases to Narrative Databases

The Narrative Compiler enables the creation of a **Narrative Database** storing **Crystallized Narrative States**, preserving causal logic, emotional weighting, and conflict structures captured at a specific moment in time.

7.2 Information Integrity and the State Hash

A **Proof of Interpretation** is defined as:

$$P = (\mathbf{s}, h)$$

where $\mathbf{s}$ is the narrative state vector and $h = H(\mathbf{s})$. When different sources report on the same event, their narrative states can be compared:

$$d(G_1, G_2) = \|e(G_1) - e(G_2)\|$$

Large divergence indicates substantial interpretive disagreement.

7.3 Applications of the Infrastructure

Media Integrity Analysis. Automated detection of narrative inconsistencies across information sources.

Geopolitical Narrative Mapping. Monitoring how different communities interpret the same international event, allowing analysts to detect emerging narrative rifts.

Economic and Corporate Intelligence. Tracking the evolution of market narratives by observing how structured events reshape strategic interpretations.

8 Reasoning over Narrative Distributions (LCM)

A text T induces a probability distribution:

$$P(N \mid T), \quad N \in \mathcal{N}$$

representing the system’s belief over possible narrative interpretations.

8.1 Inference as Distribution Transformation

While conventional LLMs predict the next token, LCM performs inference by transforming narrative distributions from $P(N_t \mid T_t)$ to $P(N_{t+1})$ upon observing new event I_t .

8.2 Handling Ambiguity and Parallel Narrative Hypotheses

The Compiler maintains multiple candidate narrative states:

$$\{(N_i, h_i, P(N_i))\}$$

tracking parallel narrative hypotheses that may later converge as additional information becomes available.

8.3 Reasoning over Causal Trajectories

Inference in LCM is modeled as a sequence of state transitions:

$$N_t \xrightarrow{F} N_{t+1}, \quad h_{t+1} = H(N_{t+1})$$

8.4 Mathematical Framework of LCM Inference

$$P(N_{t+1}) = \sum_{N_t} P(N_{t+1} \mid N_t, I_t) P(N_t)$$

LCM performs reasoning through: (1) **Trajectory Prediction** $\hat{e}(G_{t+1})$, (2) **Consistency Check** $d(\hat{e}, e_{\text{obs}})$, and (3) **Distribution Update** via Bayesian inference.

9 Human Memory and Narrative Cognition

9.1 Memory as a Compiled Distribution

Memory can be interpreted as maintaining a probability distribution:

$$P(N \mid \text{Experience})$$

Recall can be viewed as the reconstruction $N^* = \arg \max_N P(N \mid \text{Experience})$.

9.2 Semantic Primitives of Biological Storage

Human cognition prioritizes: **Key Actors**, **Causal Interpretations**, **Emotional Evaluation**, and **Outcomes**—closely resembling the structural components of the Narrative Quantification framework.

9.3 Biomimetic Interpretation

Cognitive dissonance can be interpreted geometrically as:

$$D = d(e(G_{\text{memory}}), e(G_{\text{new}}))$$

When this divergence is large, cognitive systems may attempt to reconcile the discrepancy by updating the internal narrative distribution.

10 Limitations

10.1 Subjectivity and Algorithmic Bias

The priors used in computing $P(N \mid \text{text})$ often originate from training data of the underlying models, potentially inheriting systematic biases favoring particular cultural or logical frameworks.

10.2 Computational Complexity of Graph Generation

For texts involving n events, the number of potential causal edges scales approximately as $O(n^2)$, and the number of possible graph configurations can grow combinatorially.

10.3 Cultural and Linguistic Heterogeneity

High narrative divergence between two interpretations may sometimes reflect cultural narrative differences rather than factual disagreement.

10.4 Data Sovereignty and Strategic Narrative Influence

If centralized authorities gain the power to define “standard” narrative states, alternative interpretations could be marginalized, raising concerns about strategic information control.

10.5 Semantic Loss during Compression

Subtle aspects of meaning—irony, subtext, poetic ambiguity—may be difficult to preserve when transforming raw text into structured narrative states.

11 Conclusion and Future Work

This paper introduced the theoretical framework of **Narrative Quantification**, a computational approach for formalizing the cognitive process by which humans construct meaning from fragmented information.

While modern language models estimate probabilities over sequences of tokens, the framework proposed here envisions **Large Cognitive Models (LCMs)** capable of reasoning over structured narrative representations. By elevating the fundamental unit of information from tokens to **events**, and transforming event interactions into **crystallized narrative states** through the 24TAG structural hash, the framework enables objective comparison, verification, and prediction of interpretations.

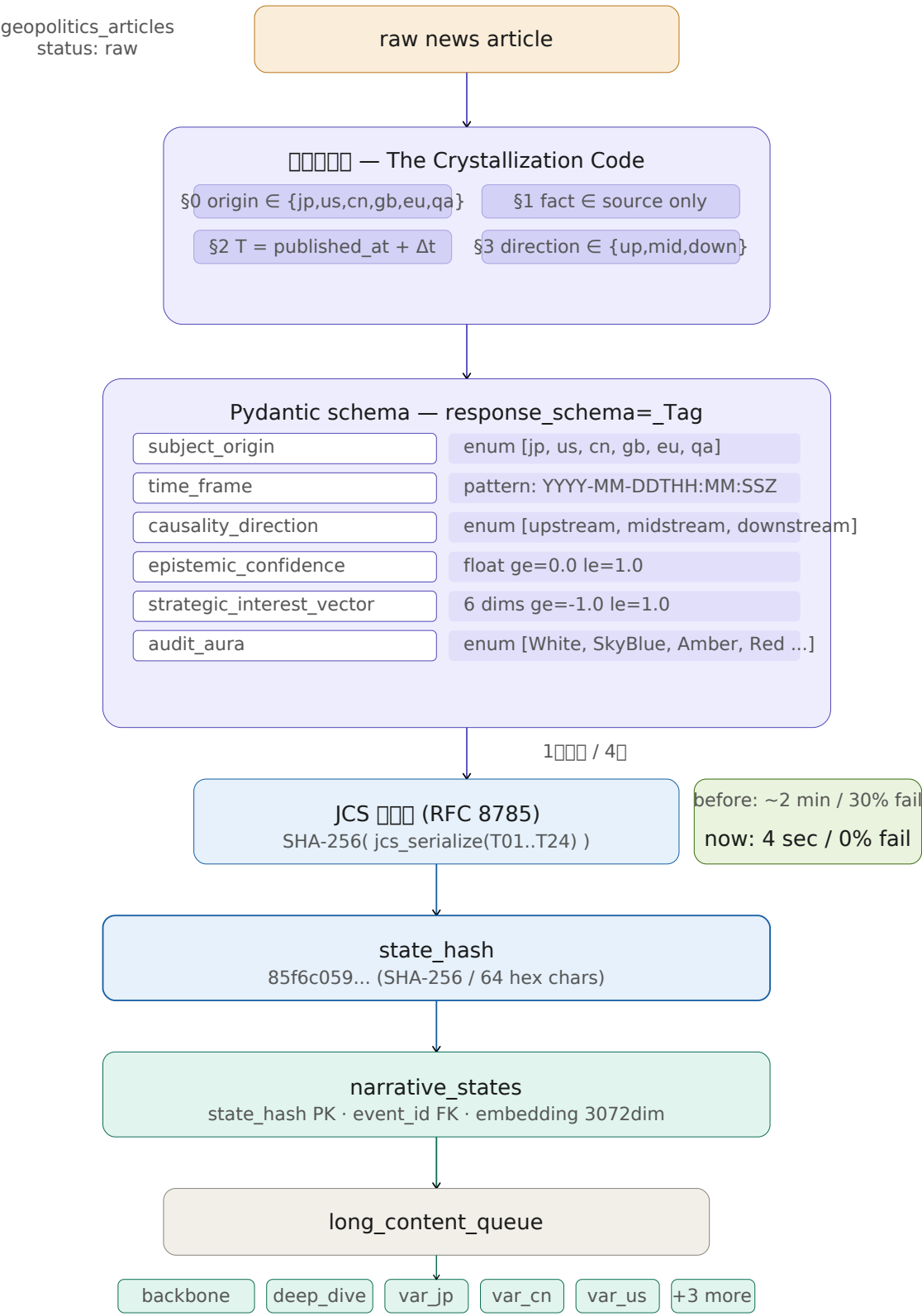
The introduction of **narrative graphs** and their embedding into a **geometric narrative manifold** allows interpretations to be analyzed within a measurable representation space. This shift suggests a transition from AI systems that merely generate fluent language to systems capable of modeling the **structural dynamics of interpretation**.

Future Work.

- **Large-scale narrative graph construction.** Scalable algorithms for constructing narrative graphs from large-scale event streams and real-world textual corpora.
- **Cross-cultural narrative modeling.** Culturally adaptive narrative embeddings across diverse linguistic and cultural contexts.
- **Deployment of narrative infrastructures.** Real-time information systems incorporating narrative hashing and narrative-space analysis for information integrity monitoring.

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ion Pipeline: An end-to-end architecture for multi-layered data purification and structural mapping. — Gemmina I

Figure 1: The Narrative Crystallization Pipeline: A deterministic measurement process that converts raw OSINT narratives into time-indexed narrative state observations.

GemminAI 24TAG 仕様書 v5.0

仕様書 — response_schema 仕様 Gemini API 仕様

JSON Schema (RFC Draft-07 / Pydantic)

Cat. I — 情報

t01 epistemic_confidence	number · ge=0.0 le=1.0	0.0 → 1.0
t02 subject_origin	enum · \$0 国	jp · us · cn · gb · eu · qa
t03 predicate_type	string · 型	declare · sanction · invest ...
t05 location	object · \$1 場所	country: string · city: null?
t06 time_frame	pattern · \$2 期間	d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}

Cat. II — 因果

t07 subject_actor	string · \$1 主体	
t08 causality_direction	enum · \$3 方向	stream · midstream · downstre
t09 strategic_interest_vector	object · 6 要素 -1.0 → 1.0	security · economy · tech · resource · ideology · environment

Cat. V — 分析Phase B

deep_dive	string · minLength: 800	1400 文字
backbone_history	string · minLength: 500	1000 文字

required fields
t01 · t02 · t03 · t05 · t06 · t07 · t08 · t09 · deep_dive · backbone_history
□ = 必須

256ビット — state_hash

state_hash = SHA-256(JCS(T01..T24)) · RFC 8785

1バイト

仕様

enum Global API	pattern 正規表現	minLength 最小長	required □ = 必須
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Narrative Crystallization: A Deterministic Framework for Measuring Global Narrative States

Supplementary Figures & Technical Specification — AIIE Protocol v5.0

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Gemmina Intelligence LLC · gemminai.com · acta-aiie.org · March 2026

Abstract

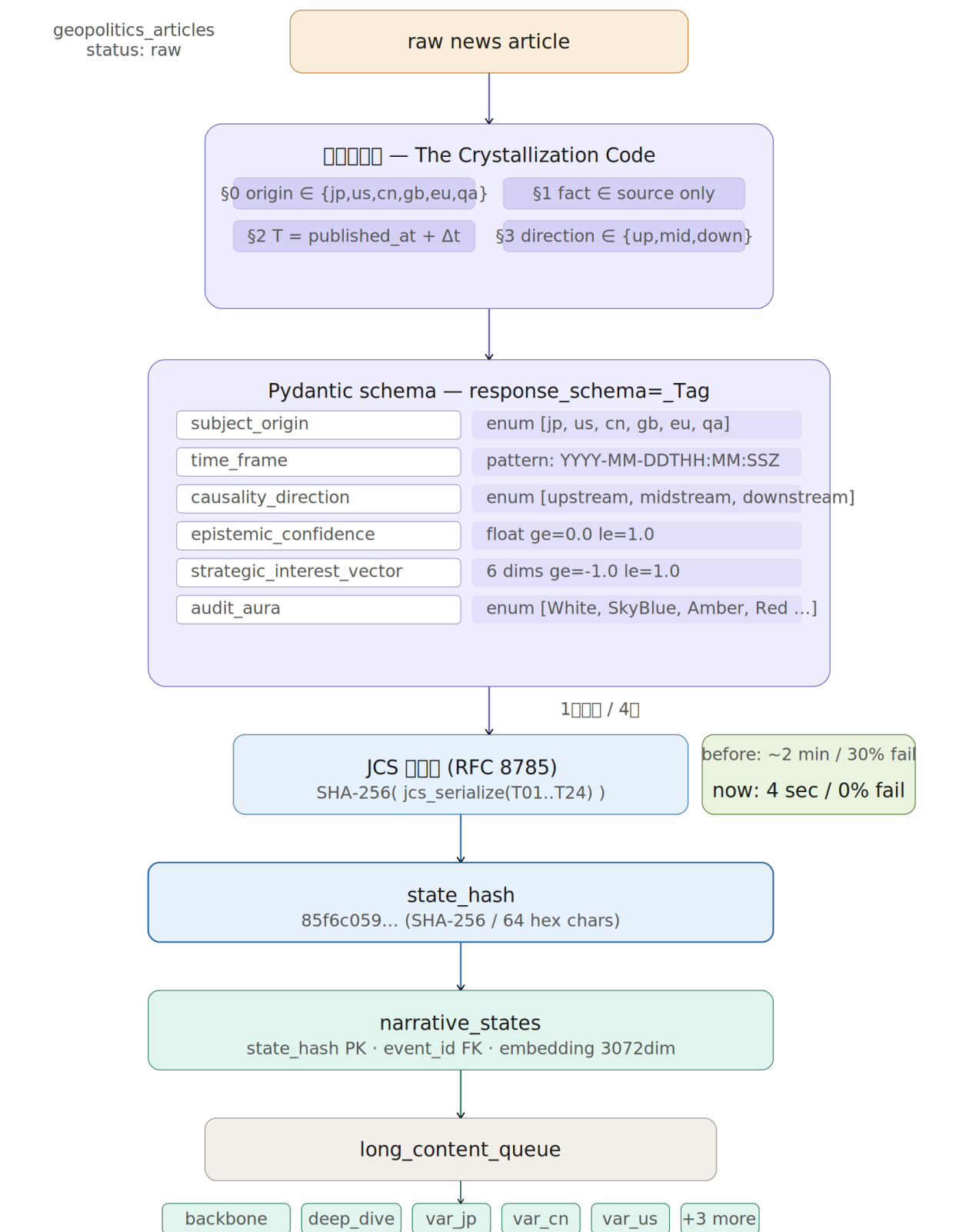
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This framework introduces a measurement layer for global narratives, enabling the construction of a time-evolving narrative state space.

Figure 1 — The Narrative Crystallization Pipeline

Insertion: Section 4 "The Narrative Compiler"



ion Pipeline: An end-to-end architecture for multi-layered data purification and structural mapping. — Gemmina Ir

Figure 1. The Narrative Crystallization Pipeline: A deterministic measurement process that converts raw OSINT narratives into time-indexed narrative state observations.

Key metrics

Metric	Before (REST + ResponseGuard)	After (SDK + Pydantic)
Processing time	~2 min / article	~4 sec / article
Schema violation rate	~30%	0%
JSON parse errors	Frequent	Physically impossible
Corpus	64 articles	64 / 64 crystallized

Figure 2 — AIIE Protocol v5.0 JSON Schema

Insertion: Section 5 "The 24TAG Taxonomy"

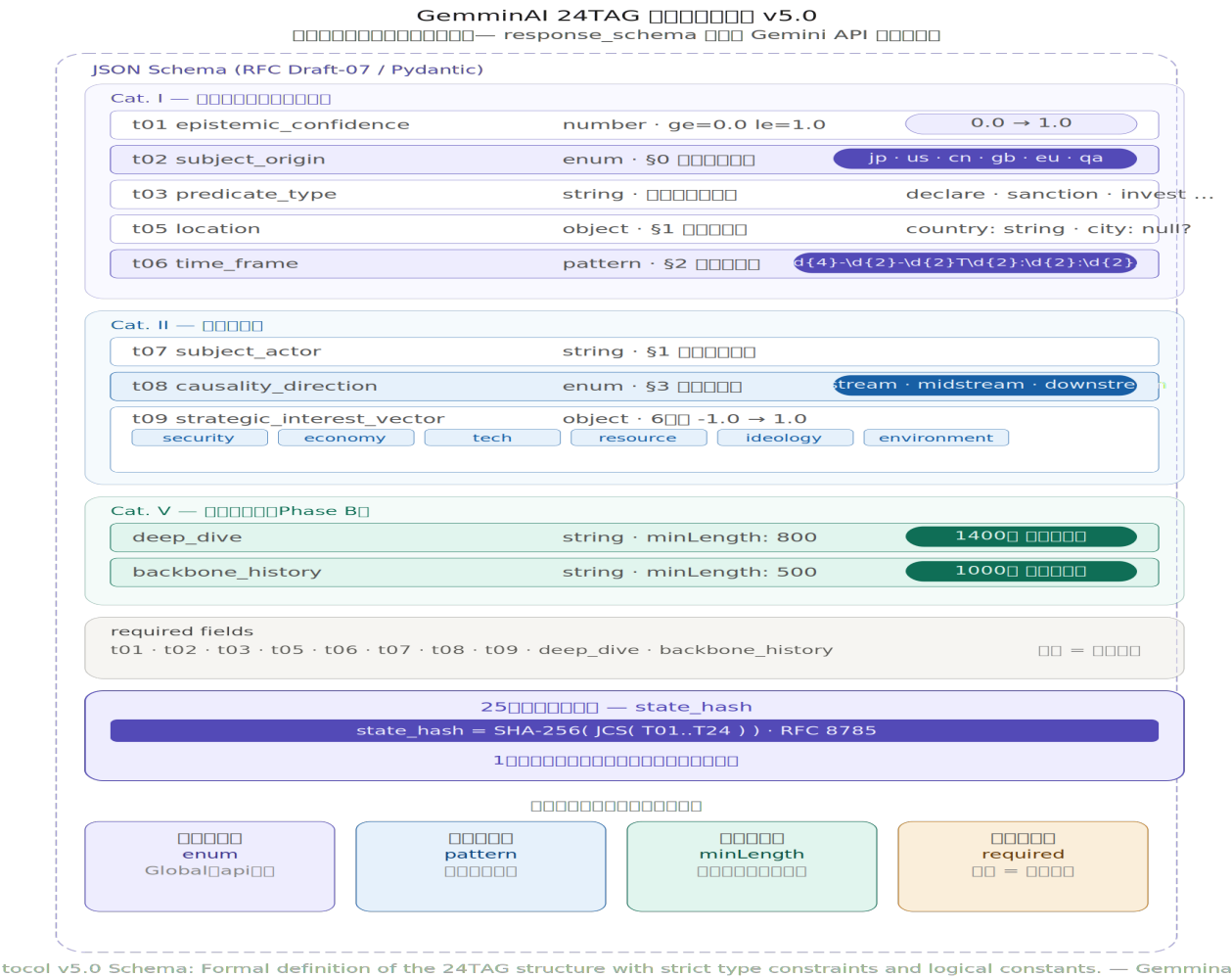


Figure 2. AIIE Protocol v5.0 JSON Schema: Formal specification for deterministic narrative states with regex-based temporal anchoring and categorical enums.

Enforcement pillars

Mechanism	Target fields	Effect
enum	T02 subject_origin, T08 causality_direction	Rejects "Global" at API level
pattern	T06 time_frame	ISO 8601 UTC/Z enforced — no timezone drift
ge / le	T01 epistemic_confidence, T09 SIV	Clamps to valid numeric range
minLength	deep_dive (800), backbone_history (500)	Prevents short-form output
required	All primary TAGs	Incomplete output = re-generation

The Crystallization Constitution (§0–§3)

\$0 subject_origin ∈ {jp, us, cn, gb, eu, qa} Determined by narrative perspective, not language. GLOBAL is forbidden. \$1 Fact ∈ Source No entity may be generated beyond what the

```
source explicitly states. Unknown location → city = null. Inference = forgery. §2 T =  
published_at + Δt time_frame must be ISO 8601 UTC/Z (19 chars). Vague terms forbidden. §3  
Direction ∈ {upstream, midstream, downstream} causality_direction is a 3-choice enum.  
"Complex correlation" is not an answer.
```

state_hash — The 25th Constant

```
state_hash = SHA-256( JCS( T01 .. T24 ) )
```

Computed via RFC 8785 (JSON Canonicalization Scheme) immediately before DB write. Any single-bit tampering produces a detectably different hash — the tamper-evident fingerprint of each narrative crystal.